

CV OF ZAKIA ZAMAN

+61-450-782-052 z.zaman@unswalumni.com Zakia Zaman Google Scholar Australian Citizen

SUMMARY

- PhD in Computer Science and Engineering, UNSW Sydney (December 2025), specialising in privacy-preserving machine learning for IoT data.
- Proven expertise in designing privacy-aware AI solutions, including LLM-based forecasting models, differential privacy, privacy-focused data encoding, and synthetic data generation, with implementations in TensorFlow, and PyTorch.
- Strong foundation in algorithms, machine learning, time series forecasting, and data privacy principles.
- Over 5 years of teaching experience in data structures, algorithms, and object-oriented programming.
- Active reviewer for IEEE Transactions on Information Forensics and Security, IEEE Transactions on Big Data, IEEE Transactions on Cognitive Communications and Networking, IEEE Transactions on Cognitive Communications and Networking, IEEE Open Journal of the Communications Society, AAAI 2026 Workshop RSD, ACM Transactions on Sensor Networks.

WORK EXPERIENCE

University of New South Wales, Sydney, Australia

Term 3 2026 (Commencing)

Casual Academic

- Accepted an offer for a casual academic (teaching) position at UNSW Sydney, commencing Term 3 2026, continuing academic engagement at UNSW following completion of the PhD.

University of New South Wales, Sydney, Australia

Jan 2026 – April 2026

Postdoctoral Writing Fellow (Part-Time)

- Led independent and collaborative research writing, including preparation of journal manuscripts, conference papers.
- Revised and strengthened existing research outputs in response to peer-review and editorial feedback.
- Synthesised prior PhD research outcomes into publishable extensions and new research directions.
- Supported dissemination of research through targeted submissions to high-impact venues.

Daffodil International University, Dhaka, Bangladesh

Apr 2015 – Apr 2021

Lecturer (Senior Scale)

Teaching Experience

- Programming and Problem-Solving (using C)(Theory & Lab), Data Structure (Theory & Lab), Object Oriented Programming (using Java) (Theory & Lab), Algorithm (Theory & Lab), Digital Electronics

Student Advising and Counselling

- Act as a Batch Advisor for more than 300 students

Final Year Project Supervision

- Research-Based Project
- Web Development Based Project

Paul Clarke & Associate Pty. Ltd. Newcastle, NSW, Australia

Dec 2009 – Dec 2010

System Support

- Design, Develop and Maintain the Office Database
- System Automation
- IT Support

University of Newcastle, Newcastle, NSW, Australia

Mar 2009 – May 2009

Research Assistant

RESEARCH EXPERIENCE

PhD Research

UNSW, Sydney, Australia

Thesis Title: Privacy Preserving Machine Learning for Internet of Things

Supervisors: Prof. Wen Hu, Prof. Sanjay Jha and Dr. Praveen Gauravaram

- Designed a privacy-preserving probabilistic data encoding technique for IoT datasets, supporting both classification and forecasting tasks. Ensures an effective trade-off between data utility and privacy when integrated into deep learning models.

- Proposed a novel framework leveraging large language models (LLMs) to enable privacy-preserving load forecasting for smart meter data, addressing privacy in downstream analysis while preserving model accuracy.
- Developed a differentially private synthetic data generation method to support privacy-aware downstream analysis of smart meter datasets, preserving key statistical properties for safe sharing and evaluation.
- Utilized TensorFlow and PyTorch to implement and evaluate models, with an emphasis on balancing strong privacy guarantees and practical performance in forecasting and analytics.
- Visualised high-dimensional data using UMAP for dimensionality reduction and Matplotlib/Seaborn for exploratory and results analysis, supporting model interpretation and evaluation across privacy-utility trade-off experiments.

Master's Thesis Research

BUET, Dhaka, Bangladesh

Thesis Title: Novel Approach for Virtual Network Function Requirement Prediction

Supervisor: Prof. Mahmuda Naznin

- Investigated predictive modeling techniques to estimate Virtual Network Function (VNF) requirements based on dynamic network traffic patterns.
- Developed and evaluated deep learning models, including CNN and LSTM, to improve accuracy in forecasting VNF scaling needs.
- Demonstrated improved performance over conventional ML methods, contributing to efficient resource provisioning in Network Function Virtualization (NFV) environments.

Undergraduate Thesis

BUET, Dhaka, Bangladesh

Thesis Title: Challenges and Issues of Adding Mobility to Wireless Sensor Network

Supervisor: Prof. Mahmuda Naznin

- Analyzed key challenges in mobile sensor networks: coverage, localization, routing, and data handling
- Evaluated distributed techniques for GPS-free localization and routing under mobility
- Proposed enhancements to existing algorithms for coverage repair and collision avoidance (e.g., Fibonacci Backoff)

EDUCATION

University of New South Wales (UNSW), Sydney, Australia	
Doctor of Philosophy - PhD, Computer Science and Engineering	2025
Bangladesh University of Engineering & Technology (BUET), Dhaka, Bangladesh	
M.Sc Engg.in Computer Science and Engineering	2020
Bangladesh University of Engineering & Technology (BUET), Dhaka, Bangladesh	
B.Sc Engg.in Computer Science and Engineering	2009

PUBLICATION (SELECTED)

- Zaman, Z., Gauravaram, P., Hassan, M., Jha, S., & Hu, W. (2026). Privacy-Preserving Machine Learning for IoT: A Cross-Paradigm Survey and Future Roadmap. *arXiv preprint arXiv:2603.13570* (under peer review).
- Zaman, Z., Gauravaram, P., Jha, S., & Hu, W. (2025). Bloom-LLM: Privacy-Preserving Large Language Model for Load Forecasting. In *Proceedings of the 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '25)*.
- Zaman, Z., Gauravaram, P., Jha, S., & Hu, W. (2025). Privacy-Preserving Smart Metering through Data Synthesis with Differential Privacy. In *Proceedings of the 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '25)*.
- Zaman, Z., Xue, W., Gauravaram, P., Hu, W., Jiang, J., & Jha, S. (2024). Privacy-Preserving Probabilistic Data Encoding for IoT Data Analysis. *IEEE Transactions on Information Forensics and Security*.
- Zaman, Z., Rahman, S., Rafsani, F., Rahman, I. R., & Naznin, M. (2023). DeepVRM: Deep Learning Based Virtual Resource Management for Energy Efficiency. *Journal of Network and Systems Management*, 31(4), 66.
- Zaman, Z., Xue, W., Gauravaram, P., Hu, W., & Jha, S. (2022, October). BloomXNOR-Net: privacy-preserving machine learning in IoT. In *Proceedings of the 13th ACM Wireless of the Students, by the Students, and for the Students Workshop*.
- Zaman, Z., Rahman, S., & Naznin, M. (2019, December). Novel approaches for vnf requirement prediction using dnn and lstm. In *2019 IEEE Global Communications Conference (GLOBECOM)*. IEEE.
- Bristi, W. R., Zaman, Z., & Sultana, N. (2019, July). Predicting imdb rating of movies by machine learning techniques. In *2019 10th international conference on computing, communication and networking technologies (ICCCNT)*. IEEE.
- Sharmin, S., & Zaman, Z. (2017, December). Spam detection in social media employing machine learning tool for text mining. In *2017 13th International Conference on signal-image technology & internet-based systems (SITIS)*. IEEE.

AWARDS

- ACM BuildSys 2025 Student Travel Grant
- Cyber Security Cooperative Research Center (CSCRC) PhD Scholarship
- Research Training Program (RTP) PhD Scholarship
- BUET Technical Scholarship

TECHNICAL COMPETENCIES

Programming Languages: C, C++, Java, Python

Tools/Framework: Keras-Tensorflow, PyTorch, Matplotlib, Seaborn, UMAP, Weka, Matlab, Cisco Packet Tracer

Web Development: ASP.Net, HTML, CSS, JavaScript

Database: Oracle, SQL

CONFERENCE (/SYMPOSIUM/WORKSHOP) ATTENDANCE/ PROFESSIONAL TRAINING (SELECTED)

- Presented two full papers at the ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys'25), Golden, Colorado, USA (2025).
Titles: “Bloom-LLM: Privacy-Preserving Large Language Model for Load Forecasting” and “Privacy-Preserving Smart Metering through Data Synthesis with Differential Privacy”.
- Poster presented at Cyber CRC Symposium 2023, held at QUT, Brisbane on July 03–04, 2023.
Title: “Synthetic Data vs. Encoded Data: Which Method to Choose in Privacy-Preserving Downstream Analysis?”
Also accepted for presentation at UNSW.AI Symposium 2023 and Australasian Computer Science Week (ACSW) 2024.
- Attended CSCRC Annual Symposium 2021–2023.
- Paper presented at ACM S3 Workshop (held in conjunction with ACM MobiCom 2022) in Sydney (16–20 October 2022).
Title: “BloomXNOR-Net: Privacy-Preserving Machine Learning in IoT.”
- Workshop on “Role of AI and Data Science in Cyber Security”, jointly organized by UNSW/Data61 on October 19, 2021.

PROFESSIONAL MEMBERSHIP

- Member(Student), IEEE
- Member(Student), ACM
- Member (Student), IEEE Women in Engineering
- Lifetime Member, Institute of Engineers, Bangladesh, (IEB)